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The HEFTCom2024 winning model: A stacked CatBoost approach for probabilistic wind and solar power forecasting

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ABSTRACT

Effective energy trading requires probabilistic forecasts to quantify uncertainty and manage financial risk. In this paper, we describe our approach, which combines separate wind and solar models using a state-of-the-art stacked CatBoost framework. The effectiveness of this method was validated in the HEFTCom2024 competition, where it was the winning entry for forecasting and trading the combined generation of a 1200 MW offshore wind and 2400 MW solar portfolio in England. Key factors contributing to our success include the use of separate wind and solar power models, the incorporation of three different weather forecast datasets, no missed submissions (benchmark fills), and effective handling of a long-lasting cable issue for the offshore wind farm. Although the main focus was on the forecasting model, we also won the trading track. This is attributed mainly to our forecast accuracy, but our trading score exceeded expectations based on the trend in trading vs. the forecasting scores of co-competitors.

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1. Introduction

As the penetration of variable renewable energy increases in a power system, it becomes more important to provide accurate forecasts.[1] Large forecast errors can lead to system imbalances and high costs for both plant owners and society. In order to deal with the inherent uncertainty, probabilistic forecasts are key.

The Hybrid Energy Forecasting and Trading Competition 2024 (HEFTCom2024) was organized by the IEEE Power and Energy Society Working Group on Energy Forecasting and Analytics. The objective was to submit day-ahead quantile forecasts and to trade the combined generation from the 1200 MW Hornsea 1 offshore wind farm and approximately 2400 MW of solar power in the East of England. There were two separate tracks, forecasting and trading, but the competitors were encouraged to participate in both.

Earlier energy time series forecasting competitions include the suites of Global Energy Forecasting Competitions (GEFComs) (Hong et al., 2016, 2019) and the Big Data Energy Analytics Laboratory (BigDEAL) (Shukla & Hong, 2024). These consist so far of three editions each, some of which were probabilistic in nature. The BigDEAL challenges and the latest iteration of the GEFCom target different aspects of load forecasting, while the two first GEFComs (2012 and 2014) also included renewable generation forecasting. Two main takeaways from the top-performing submissions of GEFCom2014 are that gradient boosting machines (GBMs) and quantile regression forests are strong models for energy forecasting purposes, especially when it comes to forecasting electrical generation, and that the combination of different forecasts (ensembling) can increase accuracy and robustness. In particular, the top-ranked submission in the wind forecasting track of GEFCom2014 (Landry et al., 2016) used

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[1] The numerical results presented in this manuscript were reproduced by the IJF reproducibility team on 25 February 2026.

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layered GBMs in a way resembling our stacked CatBoost model.

These findings were generally confirmed by Bojer and Meldgaard (2021), who analyzed six recent Kaggle forecasting competitions involving daily or weekly time series. They reported that all competitions were won by teams employing ensemble and cross-learning approaches, and that GBM or deep learning methods won the four most recent competitions.

Another relevant suite of time series forecasting competitions, although not energy-related, is the Makridakis competitions. The fifth edition (M5) focused on accuracy and featured both a deterministic and a probabilistic part (Makridakis et al., 2022). Like for HEFTCom2024, probabilistic submissions were scored with the pinball loss. Four out of the top five ranking teams in M5 used LightGBM, which is similar to the CatBoost models used here, while one team employed a long short-term memory model. The goal of the sixth and most recent Makridakis competition (Makridakis et al., 2024) was to predict the performance of a set of publicly tradable assets and optimize investment decisions to maximize returns. In this more stochastic setting, a broader set of methods ended up among the top submissions, but it was again confirmed that the practice of combining forecasts is a simple but effective way to increase accuracy.

Intense academic research has been conducted on renewable forecasting during the last decade. For short-term forecasting purposes, machine learning methods like the support vector machine and random forests have historically proven useful, but a shift towards deep learning-based approaches and various hybrid methods is noted by Tuncar et al. (2024). A plethora of deep learning architectures have been proposed, including variants and combinations of recurrent neural networks for temporal feature extraction and convolutional neural networks for spatial feature extraction; see e.g. Li et al. (2025). However, as seen above, these have not yet proven generally superior in the large forecasting competitions.

Svenska kraftnät (SVK) is the transmission system operator of Sweden. Its two main tasks are to maintain and develop the transmission grid ($\geq$220 kV) and to keep the short-term balance of the power system. During the last five years, a data science team has been established, dealing with in-house analyses, recommendations, and forecasts such as consumption, wind power, imbalance, system inertia, and power flows, with forecast horizons spanning mainly from nowcasting to two weeks. For the competition, our main focus was on the forecasting track.

An overview map of the region of interest for the competition is shown in Fig. 1. The boxes around the Hornsea 1 wind farm and photovoltaics (PV) in the East of England represent the extent of the GFS and DWD weather forecasts provided by the competition organizers. The dotted line shows the full domain of the additional MEPS dataset.

An example of quantile forecasts, trade, and actual generation for one day (i.e. one submission in the competition) is shown in Fig. 2. Note that our trade is not equal to $\hat{q}_{0.5}$, which could be considered the naive strategy. The competition was supposed to start with submissions for

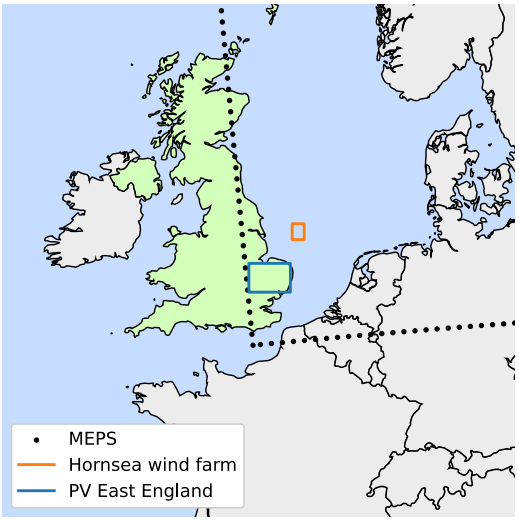


Fig. 1. Overview of the region of interest for the competition. The boxes around the Hornsea 1 wind farm and photovoltaics (PV) in the East of England represent the extent of the GFS and DWD weather forecasts. The dotted line shows the full domain of the MEPS forecasts.

market day 2024-02-01, but competitors were allowed to post forecasts before that and scores were updated daily on a leaderboard. Just a few days before the competition started, scores were suddenly poor for all competitors. It was found, and later communicated by the competition committee, that this was due to a cable problem for the Hornsea 1 wind farm that led to a limited export capacity. We handled this issue by simply putting a manual cap on the wind generation; see details in Sections 2.3 and 4. In Fig. 2, the export limitation is clearly visible after around 18:00. In order to give participants time to adjust their models, the start was delayed by 20 days and the competition period was thus 2024-02-20 to 2024-05-19.

The main objective of this text is to give a very practical description of a competition-winning model set. Thus, we deviate somewhat from the standard format of a scientific paper. A short description of the rules and scoring metrics is given in Section 1.1. Section 2 presents the data. Section 3 contains a detailed description of the main model set and includes results for the test period of 2023. More practical matters such as technical platforms, handling of missing data, and the Hornsea 1 cable problem are treated in Section 4. The paper concludes with a joint section on discussions, conclusions, and future work.

1.1. Rules and metrics

The competition consisted of two tracks, forecasting and trading, both concerning the combined production of the Hornsea 1 offshore wind farm and solar power in the East of England. Submissions were to be posted every day at 09:20 UTC for each 30-minute period in the day-ahead market (samples starting from 00:00 to 23:30 local time). In the forecasting track, probabilistic forecasts in the form of quantiles $\hat{q}_{0.1}, \hat{q}_{0.2}, \ldots, \hat{q}_{0.9}$ were to be produced. These were subsequently scored using the pinball loss; for a

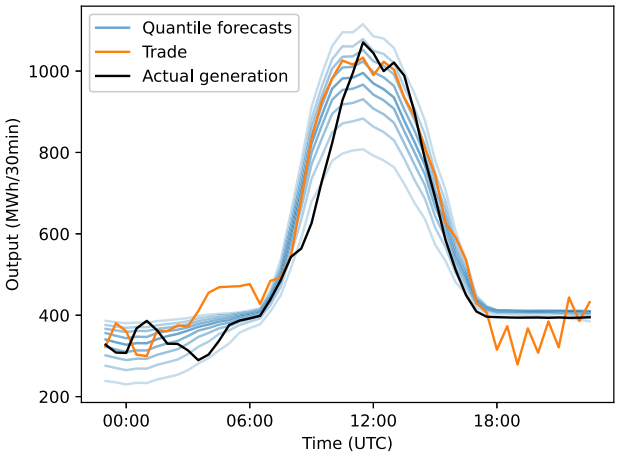


Fig. 2. Example of forecasts, our trade, and actual generation for a sunny day. The blue lines represent the nine forecast quantiles $\hat{q}_{0.1}, \ldots, \hat{q}_{0.9}$.

forecast $\hat{q}_\alpha$ of the α% quantile, and observation y, the pinball loss is given by

$$L(y, \hat{q}_\alpha) = \begin{cases} (y - \hat{q}_\alpha)\alpha & \text{if } y \geq \hat{q}_\alpha \\ (\hat{q}_\alpha - y)(1 - \alpha) & \text{if } y < \hat{q}_\alpha \end{cases} \quad (1)$$

which was averaged across all quantiles and the three-month competition period. Note that y includes curtailed energy as per instructions from the transmission system operator. This made the forecasting problem less difficult, since one would otherwise have to forecast something that depends not only on physics but also on the energy market, grid congestion, etc.

In the trading track, the objective was to post trading bids that maximized revenue. Strategic trading was not prohibited; i.e. we were allowed to bid differently than anticipated energy production. The revenue was computed from trade volume, day-ahead price (DAP), and imbalance price (IP) and summed over the three-month competition period. For each half-hour period, revenue was calculated from actual generation y and trade T as

$$Revenue = T \times DAP + (y - T) \times IP \quad (2)$$

The imbalance price IP was estimated as

$$IP = SSP - 0.07 \times (y - T) \quad (3)$$

where SSP is the actual single system price. The last term in Eq. (3) was included by the competition organizers to replicate the influence from the imbalance of Hornsea 1 and East of England PV on SSP. Note that without this term, the trading track would collapse to a classification problem (whether to trade 0 or 1800 MWh, the minimum and maximum bids according to the competition rules).

2. Data

The data were provided by the competition organizers, except where otherwise stated. Details can be found in Browell et al. (2025).

2.1. Power and trade

A bit more than three years of wind and solar power time series were provided by the competition organizers. For wind power, a corresponding time series of curtailed energy was also provided; this should be added to give the correct target. Wind power data were available with around one week delay, which limited the value of lagged wind power as a feature. Examples of 30-minute resolution time series of wind and solar production can be found in Fig. 3. In the left panels, the complementary seasonality of wind and solar can be seen. In the two-week excerpt shown to the right, the diurnal pattern of solar and the prominent synoptic (a few days) timescale of wind power variability are clearly visible (Olauson et al., 2016). Note that the unit in Fig. 3 is MWh/30 min, i.e. half the magnitude of the corresponding MW values. The installed wind power capacity of Hornsea 1 was constant at 1200 MW during the training, testing, and competition periods (2020-09 to 2024-05). The installed solar capacity increased mildly, from an estimated 2.1 GW in September 2020 to 2.6 GW in May 2024. The average capacity factor for Hornsea 1 during 2023 was 0.476, and for East of England PV it was 0.107. This implies that the average generation was markedly higher for Hornsea 1, despite the lower installed capacity.

In order to compute our target variable for the trading model (see Section 3.4), actual hybrid generation, the day-ahead price, and the single system price (SSP) were used. All these data were provided by the competition organizers.

2.2. Weather forecasts

Forecasts from three different weather models were used. We denote these GFS, DWD, and MEPS.

- Global Forecast System (GFS) from NCEP, USA. Provided by competition organizers.
- ICON-EU model from the German weather service, DWD. Provided by the competition organizers.

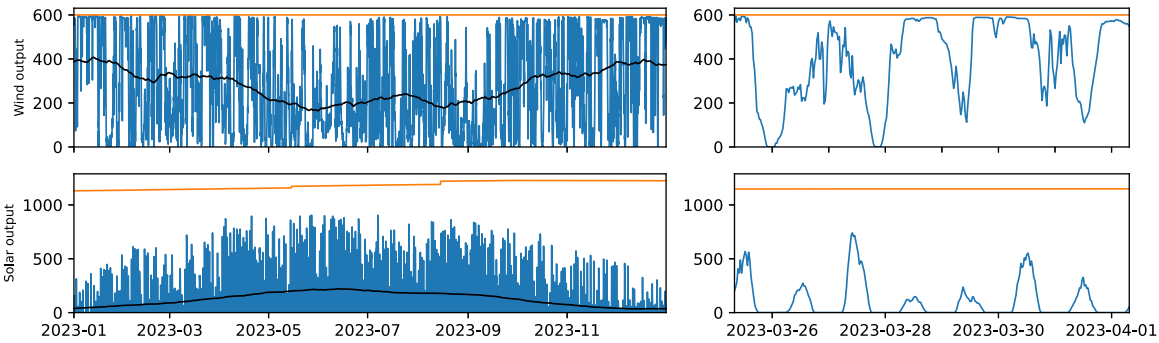


Fig. 3. Example time series of wind (upper panels) and solar power. The blue, orange, and black lines represent the output, estimated installed capacity, and 60-day rolling mean output, respectively. The left panels show data for 2023, and the panels to the right show a two-week excerpt. The unit is MWh/30 min. For wind power, curtailed energy has been added; i.e. the blue lines represent potential rather than actual generation.

- MetCoOp Ensemble Prediction System (MEPS) from the Norwegian weather service (Müller et al., 2017). Only the deterministic forecast was utilized. These data are freely available[2] and were downloaded by us for approximately the same domain as GFS and DWD forecasts.

The MEPS model is intended for the Nordic countries and, as can be seen in Fig. 1, the domain just barely covers the region of interest for the competition. We therefore suspected that the forecast quality would be poor. However, including MEPS in the model stack considerably lowered the pinball loss.

In order to train a probabilistic forecast model, it is important that the uncertainty of the input features is represented in the training data. We therefore also used day-ahead weather forecasts for training. As discussed in Section 5, this was straightforward, since the forecasts were non-overlapping; one forecast per day was submitted for the following day.

There were some missing data in the weather forecasts. For the period from 2020-09-21 to 2023-12-31, at least one of the variables was missing for 0.3%, 1.5%, and 1.8% of the time for GFS, DWD, and MEPS, respectively. In order to get robust models, this had to be handled; see Section 3.

For wind power, the following forecasted variables were provided by the competition organizers for GFS and DWD (Hornsea 1 domain; see Fig. 1):

- Wind speed and direction at 10 m and 100 m above ground
- Temperature at 2 m above ground
- Relative humidity at 2 m above ground

For MEPS, wind speed and direction at 50, 100, and 150 m height were computed from model-level heights in addition to the variables listed above. Air pressure forecasts were also included. The computation of wind at fixed heights above ground was performed in two steps. First, the time and space varying heights for the eight lowest model levels were computed; see the code in the Supplementary Material. The method is similar but not identical to that provided by the MEPS publisher.[3] Second, the variables at 50, 100, and 150 m were estimated using a cubic fit of wind speed vs. the logarithm of height above ground. With this method, the profile will be locally log-law-shaped but is forced to match exactly the values at the model-level heights.

For solar power, the following variables were provided by the competition organizers for GFS and DWD (East of England domain; see Fig. 1):

- Temperature at 2 m above ground
- Solar downward radiation
- Total cloud cover

For MEPS, only temperature and total cloud cover were available. We also found that adding all of the wind power variables to the solar model improved performance. That is, wind speed, etc. for the Hornsea 1 region were used as features for the solar power forecast in the East of England.

2.3. Export cable fault

As mentioned in the introduction, there was a cable problem at Hornsea 1 just before the competition started. Since events like this have an effect on the energy market, they need to be openly communicated as REMIT (regulation on wholesale energy market integrity and transparency) messages.[4] REMIT messages are not always static or correct, so we also kept an eye on final physical notifications, i.e. the best estimates of production submitted before gate closure, which is one hour before the settlement period.[5]

An example of the impact from the limited export capacity of Hornsea 1 can be seen in Fig. 2. Wind power production is capped at around 800 MW (400 MWh/30 min). After 18:00, solar production is zero and potential wind power production is high. This results in a constant

[2] Available at https://thredds.met.no/thredds/metno.html, accessed 2024-05-23.

[3] https://github.com/metno/NWPdocs/wiki/Examples#calculating-model-level-height, accessed 2025-05-26.

[4] https://bmrs.elexon.co.uk/remit, accessed 2024-05-27.

[5] https://bmrs.elexon.co.uk/balancing-mechanism-bmu-view, accessed 2024-05-27.

Table 1
The main and two backup model sets. The pinball loss is given for our test period, the full year 2023, but only for concurrent data (93% of timestamps where all models produced valid results). In addition, results are given for CatBoost without lagged wind power as a feature, thus making a more direct comparison with the stacked equivalent. Note that this model was not used during the competition.

Model set	Pinball loss	Coverage	Comment
1. Stacked CatBoost	28.5	100%	Main model, described in detail in Section 3.
2. CatBoost	28.4	93.2%	Same as 1, except with lagged wind power as a feature and all weather data in the same model (no stacking).
3. Reference	38.0	99.6%	Quantile regression model as suggested by the competition organizers, directly predicting hybrid power.
CatBoost no target	28.6	93.2%	Same as 2, except without lagged wind power as a feature.

output of around 400 MWh/30 min. In this case, the cap in our model was correct, which resulted in a low pinball loss. In other cases, however, wrong caps lead to high losses.

3. Models and results

In this section, the main model set is described and the results are presented. Focus is on a practical description of the models and feature engineering, but there is also a more theoretical subsection on correlation-adjusted summing. Because of the cable issue of Hornsea 1, the results from the competition period are more difficult to analyze. We therefore mostly consider results from the test period in this section. Models were trained on data from 2020-09-21 to 2022-12-31 and evaluated on data from 2023.

At a high level, three different model sets, each comprising a hybrid power model and potentially underlying wind and solar power models, were developed. The idea was to use the best hybrid power model as often as possible, and when this fails to produce a forecast, to fall back on a more robust but less performant one. The three model sets are presented in priority order in Table 1. The 'CatBoost' model is the same as 'Stacked CatBoost' except that lagged wind power was used as a feature and weather data from GFS, DWD, and MEPS were features in the same model (no stacking). The 'Reference' model is a replication of the linear quantile regression model provided as a benchmark by the competition organizers. For comparison, a 'CatBoost no target' model was also evaluated. This is the same as 'CatBoost' except that lagged wind power was not used as a feature, thus making a direct comparison with 'Stacked CatBoost' more fair.

An observant reader may note that the second priority model has a slightly lower pinball loss than the first priority model. The reason for putting the 'Stacked CatBoost' model first was that this model does not use lagged wind power observations as a feature. Due to the Hornsea 1 cable fault, we were uncertain how lagged wind power would impact the predictions.

The architecture of the 'Stacked CatBoost' model set is shown in Fig. 4. Three different wind and solar power models were trained separately, each of them using weather data from only one forecast model. Two meta-models (the wind stack and the solar stack in Fig. 4) were trained on the output from the individual forecasts. Each of the meta-models had nine sub-models for the different quantiles. Quantile forecasts from the wind and solar stacks were subsequently combined into a hybrid forecast. The trade model used hybrid quantile forecasts to predict the optimal trading.

The remainder of this section solely concerns the 'Stacked CatBoost' model set: Section 3.1 deals with CatBoost and stacking, Section 3.2 covers feature engineering, and Section 3.3 is a description of the summing of the wind and solar forecasts to obtain a hybrid forecast. The models and results for the trading track are presented separately in Section 3.4.

3.1. Stacked CatBoost models

CatBoost (Prokhorenkova et al., 2018) is a member of the gradient boosting family that, like siblings such as XGBoost and LightGBM, trains relatively fast and offers good performance (Bentéjac et al., 2021). The CatBoost regressor relies on input data in tabular format. That is, the user needs to construct X (feature matrix) and y (target vector). The idea of combining several forecast models is several decades old. There are many different ways of doing this, one of which is stacking. In the first level, several individual models are trained on the original data. In the second (and possibly subsequent) level, a meta-model is trained on the outputs from the prior forecasts (Wang et al., 2023).

As shown in Fig. 4, for each of wind and solar power, we use three CatBoost regressors as individual models and a quantile linear regression model as the meta-model. CatBoost supports the pinball as a loss function. This was a natural choice, because the pinball was used for scoring in the competition. Furthermore, CatBoost can be trained with a multi-quantile loss, i.e. trained once for all quantiles. This implies that the quantile forecasts do not have to be sorted, a simple technique that can otherwise improve performance (based on experiments with other gradient boosting implementations). After some non-extensive tuning, we settled on 200 iterations for the wind power model and 1000 iterations for solar. Except for that, the default hyperparameters were used.[6] The CatBoost models were trained twice. After the first training round, columns with zero feature importance were

[6] https://catboost.ai/en/docs/concepts/python-reference_catboostregressor, accessed 2024-05-31.

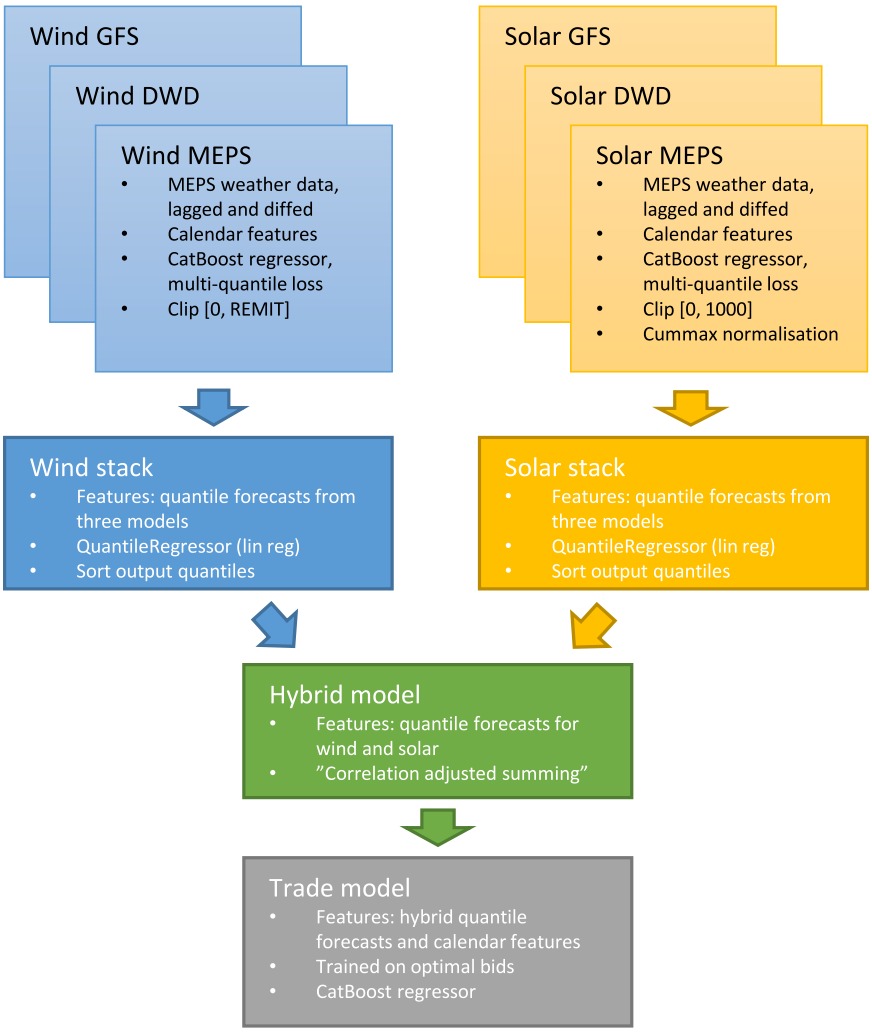


Fig. 4. Architecture of the 'Stacked CatBoost' model set. All arrows indicate sets of quantile forecasts $\hat{q}_{0.1}...\hat{q}_{0.9}$. Three different wind and solar power models were trained on data from one weather forecast model each. The outputs from these individual models were used as features for two linear quantile regression meta-models and summed to form hybrid power quantile forecasts. Outputs from this model were finally used to train the trade model.

dropped. In hindsight, this gave no performance gain. CatBoost seemed to handle redundant features without problems. For other model types, however, this strategy has helped.

A quantile linear regression model[7] was used as the meta-model. The only inputs for the $\hat{q}_\alpha$ meta-model were all nine quantile forecasts from the three individual models; i.e. X had $9 \times 3 = 27$ columns. The output was sorted so that $\hat{q}_{0.1} \leq \cdots \leq \hat{q}_{0.9}$. It can be mentioned that the same training period was used for the meta-models as for the individual models. Normally, one reserves part of the training data for the meta-models, but this was not deemed necessary, since the only difference between the individual models was the choice of the weather forecast dataset. As shown in Fig. 5, stacking improved performance considerably: the pinball losses for the full stacks (GFS+DWD+MEPS) of solar, wind, and hybrid were 6%–7% lower than the best individual models. If we look at the performance of the individual models, there are some interesting variations. GFS was clearly the worst for the solar model but slightly better than DWD for the wind power model. MEPS was by far the best dataset for wind and the hybrid, but somewhat worse than DWD for solar. A reason for the latter might be that we did not find a radiation variable for MEPS. The inclusion of MEPS (comparing GFS+DWD+MEPS with GFS+DWD) lowered the pinball loss with 7%, 8%, and 8% for solar, wind, and hybrid power, respectively.

A key advantage of our stacked models is resilience to missing data. Rows in X for the individual models with any missing data were dropped, since this could lead to unreliable predictions. Consequently, individual forecasts were not produced for these timestamps. Missing entries

[7] https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.QuantileRegressor.html, accessed 2024-05-30.

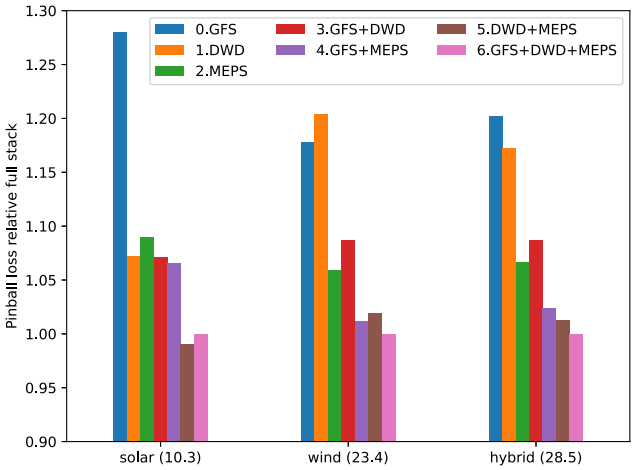


Fig. 5. Impact from stacking several CatBoost models with different weather datasets. The results are normalized with the corresponding full stack (GFS+DWD+MEPS). The pinball losses for the full stacks are given in parentheses in x-labels. For readers of the black-and-white version of the paper, the legend is numbered in the bar order (left to right).

for the meta-model were handled by filling in data, quantile by quantile, across the available individual model outputs. This method allowed us to produce a complete set of reliable forecasts for the test and competition periods even though data coverage was well below 100% for all weather models. Details of the technical implementation can be found in the Supplementary Material.

3.2. Feature engineering

In this section, all feature engineering—how raw input data are transformed into the columns in X—is described. Six calendar features were used for both the solar and wind power models: *minute*, *year*, *cos*(*hour of day*), *sin*(*hour of day*), *cos*(*month of year*), and *sin*(*month of year*).

Weather data provided by the competition organizers, as well as the additional MEPS dataset, were used. Note that the wind power variables were used for the solar models. The domain for these is around the Hornsea 1 wind farm, but they still proved valuable for model performance. Sine and cosine components were computed from wind direction (raw wind direction was not included in X). All grid points were included in X; i.e. no spatial aggregation was employed. In a post-competition analysis, the impact from using only one central grid point or a spatial mean of all grid points was investigated for the MEPS individual model. The results vary slightly with model initialization. The pinball loss for solar power was increased by around 10% (one point) and 2% (spatial mean) as compared to the model with all points. For wind power, the increase was around 2% and 1%, respectively. More elaborated techniques for reducing the dimensionality, e.g. principal component analysis (PCA), were not considered. Based on our experience, these could often reduce training time considerably, with little or no loss of performance.

For the solar models, weather features (X_w) were included in three forms:

- X_w in its raw form
- One sample difference of X_w
- X_w lagged one and two samples forwards and backwards

Table 2
Feature importance for wind power models. Each column shows results for an individual CatBoost model with its respective weather data source.

	GFS	DWD	MEPS
Wind speed	91.2%	92.2%	94.3%
Wind direction	3.8%	3.2%	1.4%
Temperature	1.0%	1.7%	1.3%
Air pressure	–	–	1.0%
Relative humidity	0.7%	0.6%	0.5%
Calendar	3.3%	2.3%	1.6%

The same weather feature processing was performed for the wind power models, except that only wind speed features were lagged. For wind power, no normalization of the target variable was employed. This was mainly because the installed capacity was constant. For solar power, the target variable was normalized with one plus the cumulative maximum output, calculated separately for each hour of the day. This removed the increasing trend due to added PV capacity and much of the diurnal seasonality.

The summed feature importance for groups of features is given in Table 2 (wind power) and Table 3 (solar power). The reason for summing is that X for each model contains in the order of 1000 columns, which becomes impractical to analyze. Feature importance was calculated with the 'prediction values change' method.[8] For each feature, this shows how much on average the prediction changes if the feature value changes.

As can be seen in Table 2, wind speed is by far the most important feature for all wind power models. For MEPS,

[8] PredictionValuesChange feature importance; see https://catboost.ai/en/docs/concepts/fstr, accessed 2024-05-28.

Table 3
Feature importance for solar power models. Each column shows results for an individual CatBoost model with its respective weather data source.

	GFS	DWD	MEPS
Radiation	52.4%	52.3%	–
Cloud cover	5.0%	7.3%	13.4%
Temperature	6.5%	5.0%	17.5%
Calendar	26.0%	23.8%	64.9%
Hornsea 1 features	10.1%	11.5%	4.1%

air pressure was included and has some non-negligible impact. For GFS and DWD, air pressure was not available (in the data provided by the organizers). Calendar features and wind direction are more important for GFS and DWD than for MEPS. This could be because the models learn how these variables relate to the wind shear (change of wind speed with height), which was not as important for MEPS, since wind speeds at several heights were available.

We now take a deeper look into wind speed features for MEPS in order to gain some insights into the importance of feature engineering (raw data, lagging or differencing) and different grid points. There are in total 24 sub-groups of wind speed features, each comprising 31 grid points for a combination of height and feature engineering. The most important sub-group is 50 mm winds lagged 1 h forward (16% of total feature importance) followed by raw 50 m winds (14%) and raw 100 m winds (9%). All but five sub-groups have feature importance above 1%. Overall, lagged features are more important than raw (64% vs. 30%) and difference features have no importance at all. All heights contribute significantly to the model, the most important being 50 m (42%) and the least important being 10 m (12%). Among the 31 grid points of the best sub-group (50 m winds lagged 1 h), many have zero or close to zero importance. For 13 grid points very evenly distributed in the grid, the importance is above 0.5%.

The feature importances of the solar power models, shown in Table 3, display more variations than for wind. The main difference is that radiation was not found (by us) in the MEPS forecasts, and consequently, the model using MEPS had to rely on other features. It is somewhat surprising that the feature importance of cloud cover is so much lower than for the calendar: 17.5% versus 64.9%. Maybe this has something to do with the weather in the UK. Temperature has a large importance for the MEPS model but considerably less for GFS and DWD. Finally, the inclusion of wind power variables from the Hornsea 1 domain had quite a large impact, in particular for GFS and DWD.

As with wind power, we now take a deeper look into the most important feature group (radiation, total importance 52.3%) for the best individual model (DWD). Since radiation is given for a single level, there are only six sub-groups. Feature importance for raw data is 16%, for one sample difference 3%, and for lags 7%, 18%, 6%, and 2% for lagging −2, −1, +1, and +2 samples, respectively. In total, lagged data are thus clearly more important than raw data. For the best sub-group (radiation lagged one sample backwards), the 20 grid points have a feature importance between 0.3% and 2.4%. Like for wind power, there is no clear geographical pattern in the importance.

3.3. Correlation-adjusted hybrid forecast

An important question, at least in theory, is how the quantile forecasts for wind and solar power should be combined to form a hybrid quantile forecast. The naive approach is $\hat{q}_{\alpha,h} = \hat{q}_{\alpha,w} + \hat{q}_{\alpha,s}$, where subscripts h, w, and s denote the hybrid, wind, and solar forecasts, respectively. Given that the wind and solar forecasts are not perfectly correlated, we assumed that the naive approach would give overly wide prediction intervals and poorer performance in terms of the pinball loss. Our team had limited time for preparation, but in a post-competition review, it was found that similar ideas have been explored in much more detail in the literature. Camal et al. (2019) used multivariate copulas to model the correlations between probabilistic forecasts of wind, solar, and hydro power in order to aggregate them correctly. Alessandrini and McCandless (2020) used the Schaake Shuffle technique to combine probabilistic wind and solar power forecasts.

Let us begin by noting that the standard deviation (σ) of the sum of two time series a and b can be expressed as

$$\sigma_{a+b} = \sqrt{\sigma_a^2 + \sigma_b^2 + 2\sigma_a\sigma_b\rho_{ab}} \tag{4}$$

where ρ_{ab} is the correlation between a and b (Ross, 2010). As an example, with $\sigma_a = \sigma_b = 1$, we have $\sigma_{a+b} = 2$ for two perfectly correlated time series and $\sigma_{a+b} = \sqrt{2}$ for two uncorrelated time series.

In analogy with Eq. (4), we define

$$\delta_{\alpha,hybrid} = \sqrt{\delta_{\alpha,w}^2 + \delta_{\alpha,s}^2 + 2\delta_{\alpha,w}\delta_{\alpha,s}\rho_{ws}} \tag{5}$$

where

$$\delta_{\alpha,x} = |\hat{q}_{\alpha,x} - \hat{q}_{0.5,x}| \text{ for } x \in [w, s] \tag{6}$$

Finally, the hybrid quantile forecasts were computed as

$$\hat{q}_{\alpha,hybrid} = \begin{cases} \hat{q}_{0.5,w} + \hat{q}_{0.5,s} & \text{if } \alpha = 0.5 \\ \hat{q}_{0.5,w} + \hat{q}_{0.5,s} - \delta_{\alpha,hybrid} & \text{if } \alpha < 0.5 \\ \hat{q}_{0.5,w} + \hat{q}_{0.5,s} + \delta_{\alpha,hybrid} & \text{if } \alpha > 0.5 \end{cases} \tag{7}$$

In hindsight, it has been possible to verify (by sampling; we are practitioners) that these intuitively defined equations were correct, at least for unbiased and normally distributed forecast errors and where the quantile forecasts for solar and wind power have correct prediction intervals. In other words, if one knows the correlation between wind and solar forecast errors, optimal quantile forecasts for hybrid power can be computed according to the equations above.

The effect of this approach is tighter prediction intervals, in particular during periods when the uncertainty in wind and solar forecasts is of similar magnitude. In our implementation (i.e. the models used for the competition), we used a time-invariant value of $\rho_{ws} = 0.6$ based on tuning. This implicitly assumes that the process is stationary, while in reality, the correlation varies depending on the weather situation, etc. Note that 0.6 is markedly higher than the actual correlation between forecast errors, which is close to zero. Using $\rho_{ws} = 0$, however, leads to an increased pinball loss. This is likely explained by the overly tight prediction intervals in the solar and wind

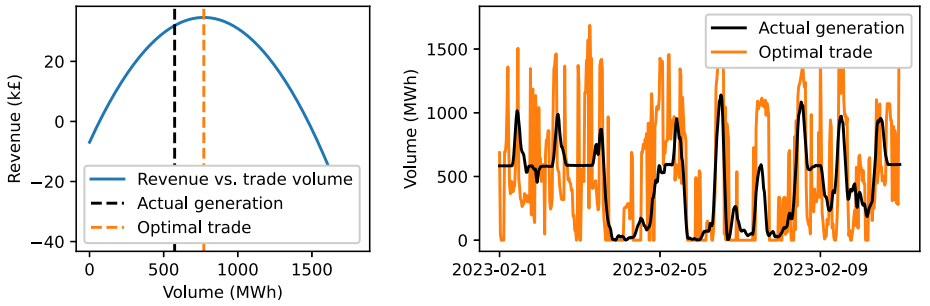


Fig. 6. Optimal trade. The left panel shows revenue vs. trade for one specific half-hour period. The right panel shows a nine-day time series of actual generation and optimal trade.

power models. Although the choice of ρ_{ws} has a relatively strong impact on the prediction interval width, the pinball loss is not affected a lot; we obtained an improvement of less than 0.1% compared to using the naive strategy $\hat{q}_{\alpha,h} = \hat{q}_{\alpha,w} + \hat{q}_{\alpha,s}$.

3.4. Trading track

Since our main focus was on the forecasting track, a relatively simple method was used for the trading model. In short, a CatBoost model with 100 iterations and an MAE loss function was trained with optimal trading as the target. As can be understood from Eq. (2), optimal trading exists for each point in time. This is visualized in Fig. 6. The left panel shows revenue versus trading volume for a specific half-hour (2020-10-21 at 06:30). With a day-ahead price (*DAP*) of 56 £/MWh and an imbalance price (*SSP*) of 28 £/MWh, it was better to sell more energy on the day-ahead market than was actually produced. The right panel shows a nine-day time series of actual generation and optimal trade. Clearly, optimal trade is more variable than the generation, which makes it difficult to predict. Looking at statistics for a longer time period (2020-09 to 2023-12), there are some patterns, however: the average *DAP* was 2.2% higher than *SSP*, and the optimal trade was 77 MWh higher than actual generation. There are also some diurnal variations of the difference between optimal trade and actual generation. In mornings and afternoons, the difference is larger than average, up to 200–300 MWh. In mid-day and evening, there is a zigzag pattern (lower difference at periods starting at integer hours than at half-hours). The latter can be seen in the example forecast in Fig. 2.

The main input features to the trading model were quantile forecasts and calendar data: *minute*, *year*, *cos*(*hour of day*), *sin*(*hour of day*), *cos*(*month of year*), and *sin*(*month of year*). The average difference *DAP* − *SSP*, calculated for each of the 48 trading periods in a day from all available data, was also added and improved performance slightly. The near-term history of prices or the optimal target was not included, since robustness (fewer required input data) was prioritized. Finally, we blended with 20% $\hat{q}_{0.5}$ forecasts. That is, the output was

$$\text{Trade model} = 0.8 \cdot \mathit{CatBoost} + 0.2 \cdot \hat{q}_{0.5} \tag{8}$$

The weight 0.2 for $\hat{q}_{0.5}$ was selected semi-manually based on revenue for the test period, 2023. In Fig. 7, the influence from using different weights is shown. Since the

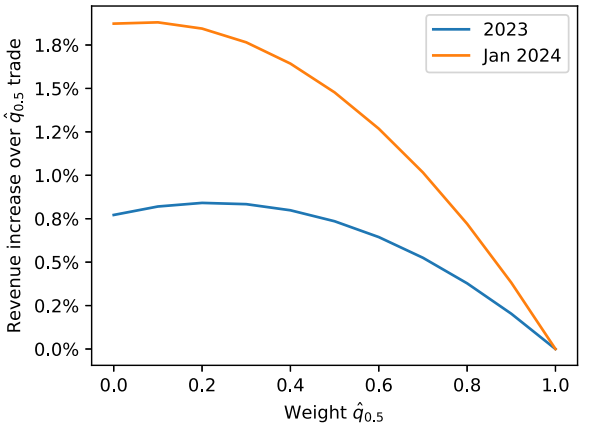


Fig. 7. Influence from changing the weight of $\hat{q}_{0.5}$ in Eq. (8) for the score in the trading track. Blending the CatBoost trade model with a small fraction of the $\hat{q}_{0.5}$ hybrid forecast improved the performance slightly.

weight was tuned for our test period (i.e. no validation set was used), the results for a different period (January 2024) are also given here. Although the improvement over a pure $\hat{q}_{0.5}$ strategy is larger for this period, the shape of the curves is similar. The optimal weight of $\hat{q}_{0.5}$ for January 2024 is 0.1 rather than 0.2.

Feature importance for the trade model is given in Fig. 8. The hybrid quantile forecasts ($\hat{q}_{0.1}, \ldots, \hat{q}_{0.9}$) have a total feature importance of 68%, and calendar features contribute 17%. The average difference *DAP* − *SSP*, calculated for each of the 48 trading periods in a day, has 15% feature importance and is thus the single most important feature. All quantiles are important, with individual contributions ranging from 4% to 11%. Interestingly, $\hat{q}_{0.5}$ (the naive trade) is one of the least important quantiles.

Trading was scored according to total revenue. Compared to a baseline strategy of submitting $\hat{q}_{0.5}$ as the bid, the algorithm outlined above increased the revenue by 0.9% during the test period. Using a perfect $\hat{q}_{0.5}$ forecast (i.e. actual generation) as trade would increase the revenue by 5.2%. An optimal trade, finally, would increase the revenue by 27.4% as compared to the baseline. During model development, 0.9% improvement over baseline

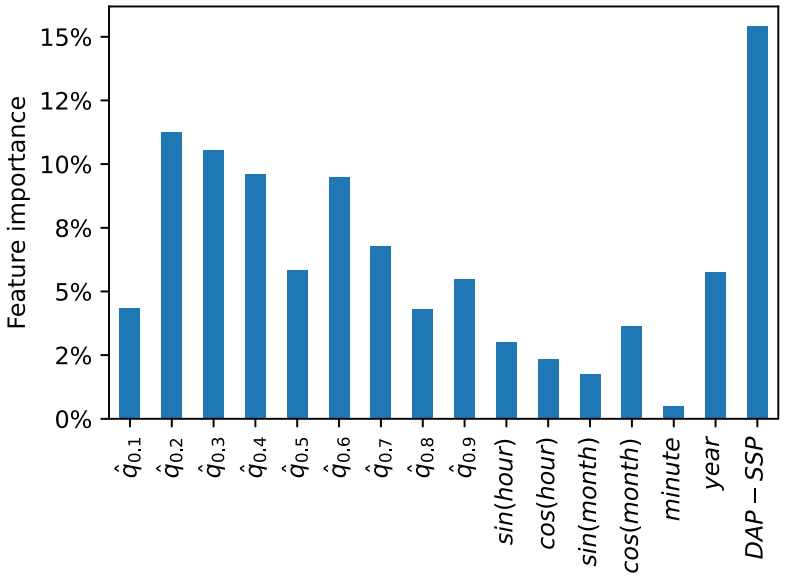


Fig. 8. Feature importance for the trading model. The hybrid quantile forecasts ($\hat{q}_{0.1}, \ldots, \hat{q}_{0.9}$) have a total feature importance of 68%, and calendar features contribute 17%. The average difference *DAP* − *SSP*, calculated for each of the 48 trading periods in a day, has 15% feature importance.

seemed futile in comparison to the potential 27.4% increase with optimal trade. Based on the competition outcome, trading is however a difficult problem, and every per mille in performance improvement counts.

During the competition period, our trading score was better than anticipated from forecasting performance; see Fig. 9. As compared to a trend calculated from the scores of other participants with good scores in both tracks, our revenue was 0.6% higher. As shown in Browell et al. (2025), our model performed consistently better than the co-competitors, except for two mistakes caused by a manual intervention (see Section 4).

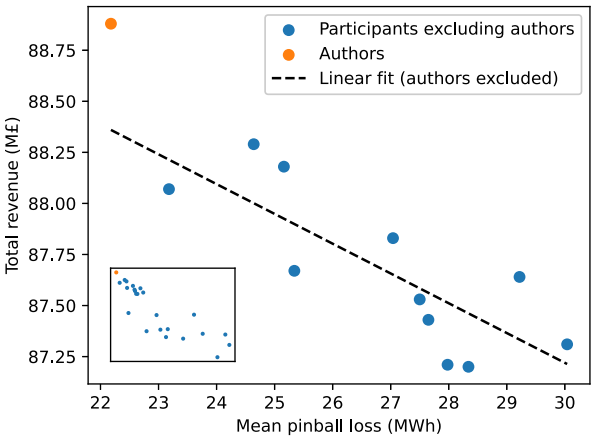


Fig. 9. Trading vs. forecasting results during the competition period for participants with good scores in both tracks (at least £87M in total revenue and less than 30.1 MWh in pinball loss). Results for all participants are embedded in the lower left. There is a distinct group of 12 teams with good scores in both tracks.

4. Practical aspects

Since we had limited time to prepare for the competition, and in particular very little time for testing the online submissions of forecasts, a belt-and-braces approach was taken. Missing a submission leads to forecasts being filled by a benchmark model run by the competition organizers. During our test period of 2023, our implementation of the benchmark model was 32% worse than the main model (Table 1); missing a few submissions was thus not thought to be critical. Due to the Hornsea 1 cable problem, which was not handled in the benchmark model, the pinball loss was 142% higher for the benchmark than our model during the competition period. Avoiding missed submissions was thus very important for the final ranking.

Model development and training were mainly performed on laptop computers, although a high-capacity CPU server was used for some tuning. Models were not re-trained during the competition period, but some alterations were necessary to deal with the Hornsea 1 cable issue; more on that below. Code, data, and the trained models were uploaded to a cloud platform where three different shell scripts ran:

- 05:00 UTC – Download MEPS 00-forecast (the weather forecast initiated at hour 00) and check for new REMIT messages (if any, send an email alert).
- 08:45 UTC – Download data from the competition API and MEPS 06-forecast. If the MEPS forecast was not ready, the 00-forecast downloaded earlier was used.
- 09:00 UTC – Run the models and submit predictions and trade. If submissions fail (errors happened a few times), sleep for one minute and retry. If submissions fail several times, send an alert. This happened one or two times and we managed to manually intervene before the submission deadline of 09:20 UTC.

The CatBoost regressor can handle missing data in X. However, this can be quite risky unless extensive testing is performed beforehand. Predictions might be significantly off if important features are missing. We thus chose to drop rows of X with any missing values. The individual models used only weather data, calendar features, and, for the solar models, solar power observations for normalization. Calendar features are naturally never missing, and solar power was forward-filled indefinitely (the impact from missing a few days of data would be minimal). Consequently, only weather data were critical for the models to function well. With three models in the stack using different weather, the final models became very robust.

As described in the introduction, the export limitation of the Hornsea 1 farm was handled by clipping the wind power forecast, i.e. a form of post-processing. This was straightforward to implement, since wind and solar power were forecasted separately. From a practical point of view, the maximum output as estimated from REMIT messages, and preliminary production data (Section 2.3) were manually filled in a text file and parsed by the model. In hindsight, it would have been better to fully automate this handling. During the last weeks of the competition, two mistakes led to high errors that nearly cost us the victory.

Training and deploying a few different model sets (see Section 3) can be valuable for several reasons. The main advantage is that one can use a model with high performance but high demands on input data as a first priority. If this fails, one can automatically fall back on simpler models. The CatBoost model, which uses a lagged target as a feature, had the best performance and was originally placed first in the priority list. Because of the export cable issue at Hornsea 1 and varying available capacity, we were uncertain how this might impact the model output and chose to put Stacked CatBoost, which does not use wind power observations as a feature, first. It can be noted that during the competition period, our implementation of the reference model (linear quantile regression, placed last in model order) failed quite often. This was because new weather data were outside the range of the corresponding training data, which was not accepted.

5. Concluding discussion

The conclusion from the M5 competition (Januschowski et al., 2022) seems to hold: tree-based algorithms have not yet been surpassed by deep learning when it comes to time series forecasting (Browell et al., 2025). Or at least, practitioners' implementations of deep learning methods are not truly successful. We also confirmed the results from many previous studies that several different weather forecasts are beneficial for variable renewable energy forecasting. Here, three individual CatBoost models trained on different weather data were combined by a linear regression meta model. For performance (pinball loss), this was actually very similar to using all weather data in one CatBoost model. This contrasts with the results for our operational models for Sweden, where model performance improved by stacking. A likely explanation is that the regions of interest for the competition were quite small. Hence, there were not too many weather data grid points, and the model could handle data from three different weather datasets at once without problems.

An interesting observation was that weather data from the Hornsea 1 region clearly improved the performance for the solar power model. Looking at feature importance, the MEPS solar power model was very different from the GFS and DWD. Since solar radiation was not available, the model had to learn from cloud cover and calendar data. The latter were by far the most prominent feature, with an importance of 65%. Interestingly, temperature was more important than cloud cover for the MEPS model. One conclusion is that it is not always easy to understand and interpret how a machine learning model learns the relationships between input features and the target variable. One has to experiment, try different features, and often redo experiments, since the results depend on model architecture and other features being added or dropped.

The setup of the competition was to submit day-ahead forecasts every day. This was in a way simpler than real-life operations where forecasts are often overlapping, e.g. creating a three-day forecast every six hours. Overlapping forecasts pose practical challenges in terms of building and training X while respecting data availability for each issue time. For a potential follow-up HEFTCom, we think it would be interesting to include this additional complexity and learn how different teams handle it. Another suggestion is to choose wind and solar areas with more rapidly increasing installed capacity over time, thus putting more emphasis on target normalization.

During model development, many ideas were tested and discarded. Other boosting algorithms and a multilayer perceptron with a custom (pinball) loss were found to be inferior to CatBoost. More advanced normalization methods for the solar target were also tried, but these provided zero or negative value. This is likely related to a very slowly increasing installed solar capacity. There were also many ideas that were never evaluated, mostly due to time constraints. A non-exclusive list follows below.

- Train separate models for sub-farms (wind power) and sub-areas (solar power).
- Use ensemble weather forecasts. We did some preliminary tests with incomplete data without success, but logically this information should be useful for quantile forecasts.
- More and better features for the trading model and perhaps a more bottom-up approach should be tried using price forecasts.
- Build deeper and/or wider model stacks, e.g. using different training periods, machine learning models, and features.
- Hyperparameter tuning. We did not tune the CatBoost models, except for the number of iterations.
- Post-processing of the quantile forecasts. Our more recent experiments for Sweden showed that enforcing a normal or skew normal distribution of the quantiles can improve the results as compared to only sorting them.

- More weather datasets and more variables—in particular, solar variables for the correct region. Recall that we used some weather data from the Hornsea 1 region as input to the solar model.
- Automatize REMIT handling. This is something we neglected in Sweden so far but will now prioritize.

As to our own goals for this competition (Section 1), we conclude that (i) quantile forecasting is not as complicated as anticipated, (ii) our model architectures and feature engineering seem to be competitive, and (iii) it was fun and we learned important lessons that can be implemented in operational models.

CRediT authorship contribution statement

Jon Olauson: Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Olle Viotti:** Writing – review & editing, Software, Methodology, Data curation. **Jakob Huss:** Writing – review & editing, Software, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

All code and input data for reproducing the results are available on Gitlab: https://gitlab.com/svenska-kraftnat/heftcom2024.

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